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* TCP / IP Stack

- The TCP/IP model has 4 layers (unlike OSI's 7.)

TCP/IP Layer	Maps to OSI	Key Protocols
1) Application	Application + Presentation + Session	HTTP, DNS, SMTP, FTP, SSH
2) Transport	Transport	TCP, UDP
3) Internet	Network	IP, ICMP, ARP
4) Network Access	Data Link + Physical	Ethernet, Wi-Fi, MAC

* Why does this matter for DFIR?

- Every ~~for~~ forensics artefact — a PCAP, a log entry, a connection record — is tagged with IP addresses and ports from this stack. You need to read them like a native language.

* IPv4 Addressing — Core Concepts

- An IPv4 address is 32 bits, written as 4 octets:
  - 192.168.1.100

    192 → Network Part
    168 → Network Part
    1 → Host Part
    100 → Host Part

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* IPv4 Classes

Class	Range	Default Subnet	Used For
A	1-126	/8 (255.0.0.0)	Large Networks
B	128-191	/16 (255.255.0.0)	Medium Networks
C	192-223	/24 (255.255.255.0)	Small Networks

* Private IP ranges

- 10.0.0.0 /8 — Class A private
- 172.16.0.0/12 — Class B private
- 192.168.0.0/16 — Class C private
- 127.0.0.1 — Loopback (Localhost)

* DFIR Tip

- When you see a private IP in a PCAP or log, that traffic stayed inside the network. A public IP means it crossed the internet. "This distinction matters enormously in investigations."

* Subnetting — Just The Logic

- A subnet mask tells you: which bits belong to the network, which to the host.

IP : 192.168.1.100 → 11000000.10101000.00000001.{01100100}

Mask : 255.255.255.0 → 11111111.11111111.11111111.{00000000}

← Network → ← Host →

* CIDR Notation

- It just counts the 1-bits in the mask:
  - /24 = 255.255.255.0 → 256 addresses, 254 usable
  - /16 = 255.255.0.0 → 65536 addresses
  - /8 = 255.0.0.0 → 16777216 addresses

* For any subnet, you need to find:

1) Network Address — First address (all host bits = 0)

2) Broadcast Address — Last address (all host bits = 1)

3) Usable host range — Everything Between

4) Number of hosts — $2^{(\text{host bits})} - 2$